Design a 3D E-Commerce Platform Architecture on AWS

PROJECT

PRESENTATION

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BUILDING THE FUTURE OF 3D E-COMMERCE

FROM CONCEPT TO REALITY

Introduction

We are a startup team working on a next-generation e-commerce platform that transforms the way people shop online. Instead of viewing static images, our users can explore and interact with realistic 3D models of products — from furniture and electronics to fashion items — before making a purchase.

Our **vision** is to create an immersive and intuitive shopping experience that bridges the gap between online and in-store interaction.

Our Goals

To bring this vision to life, we aim to build a platform that:

- Delivers a fast and smooth 3D experience across devices
- Supports millions of users worldwide without performance loss
- Ensures high availability and uninterrupted access
- Maintains strong security and data protection standards
- Remains cost-efficient as the platform scales

OUR APPROACH

TO ACHIEVE THESE GOALS, WE ARE DESIGNING A CLOUD-NATIVE ARCHITECTURE USING AWS SERVICES. THIS APPROACH ALLOWS US TO BUILD A FLEXIBLE, SCALABLE, AND RELIABLE INFRASTRUCTURE THAT CAN GROW TOGETHER WITH OUR BUSINESS AND USER BASE.

01 **Service Choices and Justifications**

02 **Architecture Explanation**

03 **Design Trade-offs & Challenges**

THE IDEATION JOURNEY

BRAINSTORMING TO BRILLIANCE

01 High Availability

Route 53,
Multi-AZ RDS,
Amazon S3,
CloudFront

02 Scalability

AWS Lambda,
DynamoDB,
ElastiCache

05 Monitoring & Observability

Amazon CloudWatch,
AWS Trusted Advisor

03 Performance

Amazon CloudFront,
Amazon S3,
RDS Read Replicas,
ElastiCache

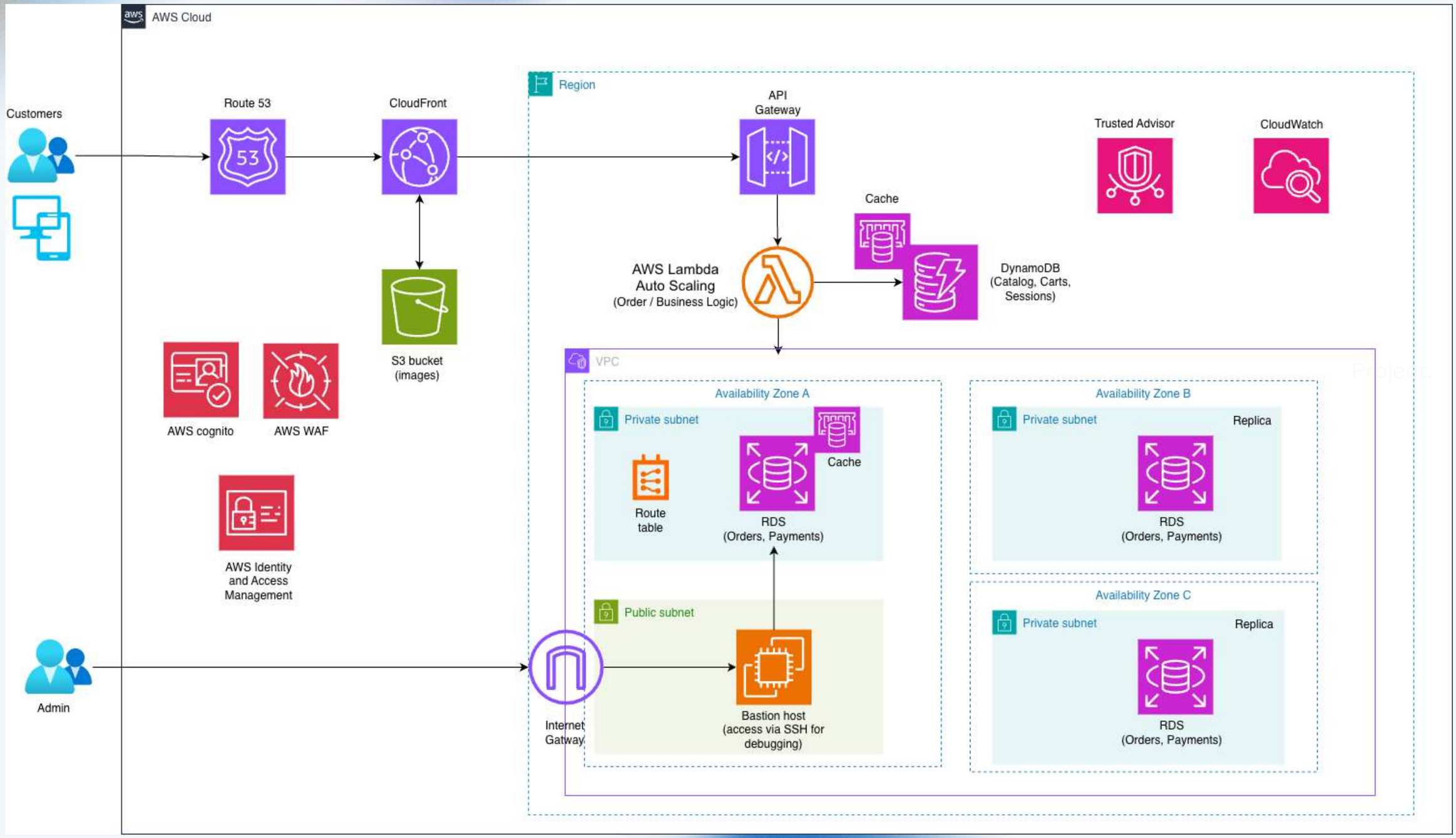
04 Security

AWS IAM,
Amazon Cognito,
AWS WAF,
AmazonVPC,
Security Groups

06 Cost Optimization

AWS Lambda,
S3 Intelligent-Tiering,
AWS Trusted Advisor

3D ARCHITECTURE DIAGRAM



DESIGN TRADE-OFFS & CHALLENGES

A EC2 vs. Lambda

B RDS vs. DynamoDB

C ElastiCache Placement

D Security vs. Performance

E 3D Asset Delivery

F Global Users & Latency

E Monitoring, Cost and Optimization

Design Trade-offs & Challenges

Strategies to Optimize Performance & Cost

Lambda-First Approach

Lambda: Auto-scaling, Time Limits | **EC2:** Debugging Only

Solution

Use **Lambda** for processing tasks with auto-scalings.
Use EC2 only for debugging purposes.

RDS & DynamoDB Sync

RDS: Transactional Integrity | **DynamoDB:** High-Scale, Fast Access

Solution

Use **RDS** for critical data & DynmooDB for sessions/3D data. Implement data sync mechanisms to keep them consistent.

3D Asset Delivery

Large 3D Models, High Bandwidth

Solution

Use S3 + CloudFront CDN.

Optimize caching to ensure fresh 3D asset delivery.

Global Deployment Strategy

Serving Millions Worldwide

Design Trade-offs & Challenges

Strategies for Caching, Security, & Optimization

Caching Strategy (ElastiCache Placement)

Challenge: Reducing database load while maintaining data freshness.

Amazon ElastiCache improves speed
But introduces:

TTL (Time-to-Live): Automatically expire outdated

Selective caching: Avoid caching sensitive transactional data

Architecture Placement: ElastiCache sits between **Lambda** and databases, acting as aperformance layer.

Solution

Cache-aside pattern:
Lambda checks cache first → fallback to DB

Cache only:
Frequently accessed, non-critical data (product views, sessions)

Architecture Decision

ElastiCache between Lambda & DBs for speed & freshness.

Security vs. Performance

Challenge: Adding strong security without increasing latency

Security layers: AWS WAF

- **Amazon Cognito**
- AWS Identity and Access Management

Resolution Strategies

Edge protection: Use WAF with CloudFront to block threats early

Token-based auth: Cognito with JWT tokens reduces repeated auth calls

Least privilege: IAM roles for secure service-to-service communication

Monitoring, Cost, and Optimization

Challenge: Maintaining visibility and controlling costs at scale

More monitoring = better insights But adds operational overhead

Architecture Decision

Built-in **ovservability** ensures proactive scaling and cost control.

THE JOURNEY CONTINUES...

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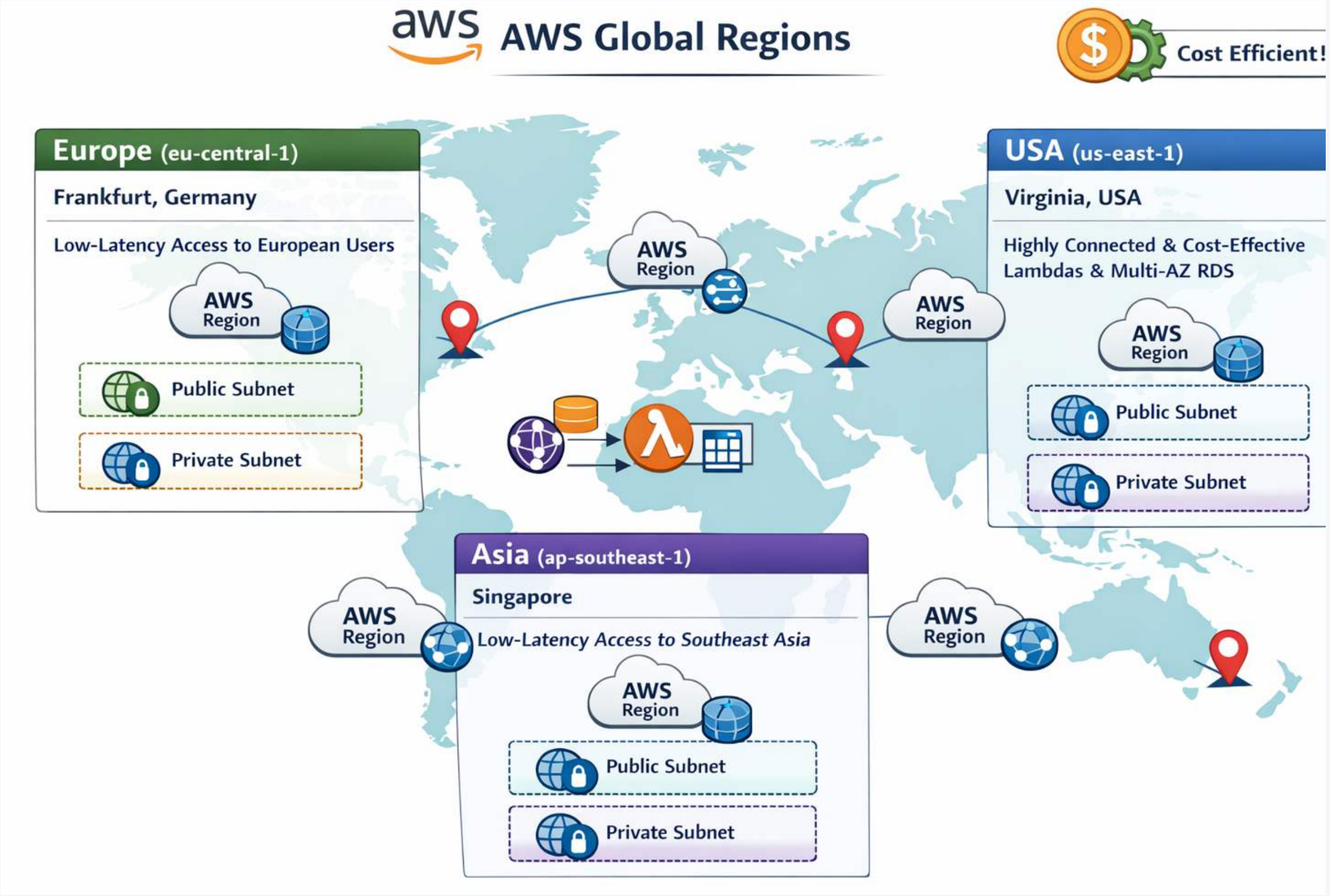
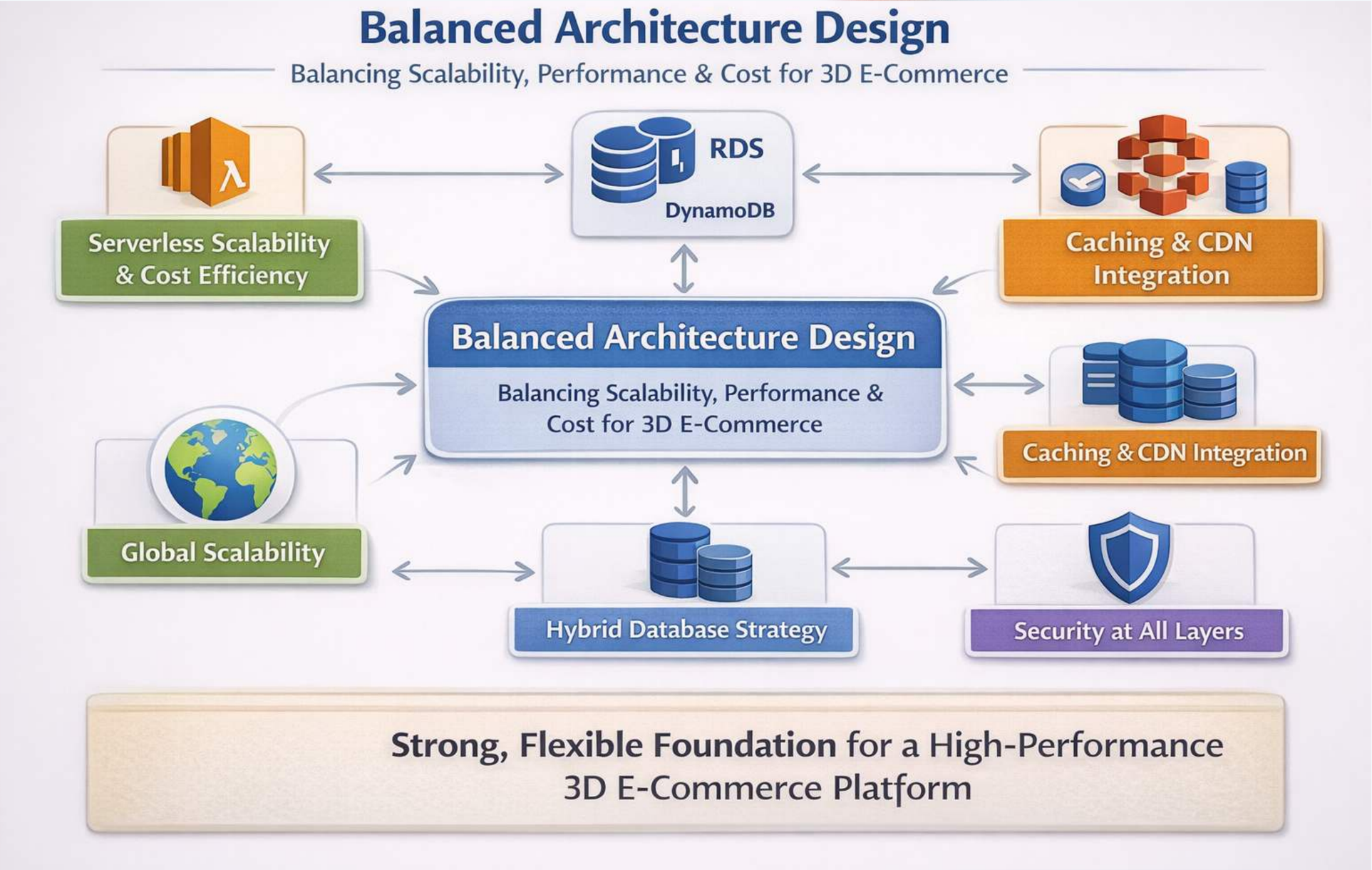


CONCLUSION


Balanced Architecture Design
Balancing Scalability, Performance & Cost for 3D E-Commerce
RDS
DynamoDB
Serverless Scalability & Cost Efficiency
Caching & CDN Integration
Balanced Architecture Design
Balancing Scalability, Performance & Cost for 3D E-Commerce
Caching & CDN Integration
Global Scalability
Hybrid Database Strategy
Security at All Layers
Strong, Flexible Foundation for a High-Performance 3D E-Commerce Platform

THANK YOU